```
/*=========================================================
                IMPORT RAW DATA
=========================================================*/

proc import
  datafile="/home/u64383806/BCMA_CLINICAL SAS..xlsx"
  out=work.BCMA_RAW
  dbms=xlsx
  replace;
  getnames=yes;
run;


/*=========================================================
                RAW DATA QC CHECK
=========================================================*/


/* Variable overview */

proc contents data=work.BCMA_RAW VARNUM;
  title "BCMA_RAW DATA OVERVIEW";
run;

/* First 50 observations */

proc print data=work.BCMA_RAW(obs=50);
  title "BCMA_RAW First Ten OBS";
run;

/* Numeric summary */

proc means data=work.BCMA_RAW mean median mode max min n nmiss;
  title "BCMA_RAW First NUMERIC SUMMARY";
run;

/* Character frequency check */

proc freq data=work.BCMA_RAW;
  tables  Chemotherapy
          Sex
         'TNM Stage'n
         'Immunohistochemistry Subtype'n
         'PAM50 subtype'n
         'Race Category'n;
   TITLE "Derived Variable Frequencies";
RUN;


/*=========================================================
            DUPLICATE SUBJECT CHECK
=========================================================*/
```

```
proc sql;
select count(*) as total,
count(distinct 'Patient ID'n) as unique_subjects
from work.BCMA_RAW;
quit;


/*=========================================================
                REMOVE DUPLICATES
=========================================================*/


proc sort data = work.BCMA_RAW out=work.BCMA_RAW_1 nodupkey;
      by 'Patient ID'n;
run;


/*=========================================================
                DEMOGRAPHICS DOMAIN
                        DM
=========================================================*/


data DM;
    set work.BCMA_RAW_1   (rename=(Sex=sex_raw));


    attrib
          STUDYID length=$20          label="Study Identifier "
           DOMAIN length=$4           label="Domain Abbrivative"
           USUBJID length=$40        label="Unique Subject Identifier"
           SUBJID length=$30          label="Subject Identifier"
           AGE length=$8              label="Age Units"
           SEX length=$4              label="SEX"
           RACE length=$20            label="RACE"
           ETHNIC length=$30          label="Ethnicity"
           COUNTRY length=$20          label="Country"
           ARM length=$40             label="Descritption of Planned Arm"
           ARMCD length=$20           label="Planned Arm Code"
           DTHFL length=$5            label="Subject Death Flag";


    STUDYID = "brca_smc_2018";
    DOMAIN  = "DM";
    SUBJID  = 'PATIENT ID'n;
    USUBJID = cats('STUDY ID'n, "-", SUBJID);
    AGE     = 'DIAGNOSIS AGE'n;


    if upcase(strip(sex_raw)) = "FEMALE" then SEX = "F";
    else if upcase(strip(sex_raw)) = "MALE" then SEX = "M";


    COUNTRY= "Korea";
```

```
    RACE='Race Category'N;
    ETHNIC ="NOT REPORTED";


    if CHEMOTHERAPY="Yes" then do;
        ARM="CHEMOTHERAPY";
        ARMCD="CHEMO";
    end;

    else do;
        ARM="NO CHEMOTHERAPY";
        ARMCD="NOCHEMO";
    end;


    DTHFL = "N";


    keep STUDYID DOMAIN USUBJID SUBJID AGE SEX RACE COUNTRY ARM ARMCD DTHFL;
    title"Demographics Domain";
run;
```

```
proc print data=dm;
run;


/*=========================================================
                DM QC CHECK
=========================================================*/
```

```
proc contents data=dm varnum ;
title "DM DOMAIN OVERVIEW" ;
run;
```

```
proc freq data = dm;
 tables
        AGE
        SEX
        RACE;

    title"DM DOMAIN FREQUENCY CHECK";
run;


/*=========================================================
                MEDICAL HISTORY DOMAIN
                        MH
=========================================================*/
```

```
data MH;
    set work.BCMA_RAW_1;

    attrib
```

```
        STUDYID length=$20  label="Study Identifier"
        DOMAIN  length=$2   label="Domain Abbreviation"
        USUBJID length=$40  label="Unique Subject Identifier"
        SUBJID length=$30   label="Subject Identifier"
        MHSEQ   length=8    label="Medical History Sequence Number"
        MHTERM  length=$100 label="Reported Medical History Term"
        MHCAT   length=$40  label="Category for Medical History"
        MHSCAT  length=$40  label="Subcategory for Medical History";

        STUDYID = "brca_smc_2018";
        DOMAIN = "MH";
        SUBJID  = 'PATIENT ID'n;
        USUBJID = cats('STUDY ID'n, "-", SUBJID);


 MHSEQ + 1;
 MHTERM = 'Cancer Type'n;
 MHSCAT = 'Cancer Type Detailed'n;
 MHCAT= "Oncology";




        keep STUDYID DOMAIN USUBJID SUBJID MHSEQ MHTERM MHCAT MHSCAT;
        title "MEDICAL HISTORY";
run;

proc print data= MH;
run;


/*=========================================================
                MH QC CHECK
=========================================================*/


proc contents data= MH varnum;
  title "MH OVERVIEW";
run;

proc freq data= MH ;
     tables  MHTERM
            MHSCAT
            MHCAT;
     title "MH DOMAIN FREQUENCY CHECK";

run;


/*=========================================================
                EXPOSURE DOMAIN
                        EX
=========================================================*/
```

```
 data EX;
    set work.BCMA_RAW_1;

    attrib
        STUDYID length=$20  label="Study Identifier"
        DOMAIN  length=$2   label="Domain Abbreviation"
        USUBJID length=$40  label="Unique Subject Identifier"
        SUBJID length=$30   label="Subject Identifier"
        EXSEQ   length=8    label="Exposure Sequence Number"
        EXTRT   length=$50  label="Name of Treatment"
        EXDOSE  length=$20  label="Dose Description"
        EXCAT   length=$40  label="Exposure Category";


    STUDYID = "brca_smc_2018";

    DOMAIN = "EX";

    SUBJID = 'PATIENT ID'n;

    USUBJID = cats(STUDYID,"-",SUBJID);


    EXSEQ + 1;


    EXTRT = "Chemotherapy";

    EXDOSE = "NOT REPORTED";


    EXCAT = 'Subtype Consensus'n;


    keep STUDYID DOMAIN USUBJID EXSEQ EXTRT EXDOSE EXCAT;

    title "EXPOSURE DOMAIN";

run;
```

```
proc print data=EX;
run;

/*=========================================================
                EX QC CHECK
=========================================================*/
```

```
proc contents data = EX varnum;
title "EX OVERVIEW";
run;
```

```
proc freq data = EX;
  tables
       EXCAT;
```

```
    title"EX DOMAIN FREQUENCY CHECK";
        run;


  /*==========================================================
                 LAB DOMAIN
                         LB
  ==========================================================*/


data LB;
    set work.BCMA_RAW_1;

    attrib
        STUDYID length=$20   label="Study Identifier"
        DOMAIN  length=$2    label="Domain Abbreviation"
        USUBJID length=$40   label="Unique Subject Identifier"
        SUBJID  length=$30   label="Subject Identifier"
        LBSEQ   length=8     label="Laboratory Sequence Number"
        LBTEST  length=$60   label="Laboratory Test Name"
        LBORRES length=$40   label="Result or Finding in Original Units"
        LBCAT   length=$40   label="Laboratory Category";


    STUDYID = "brca_smc_2018";

    DOMAIN = "LB";

    SUBJID = 'PATIENT ID'n;

    USUBJID = cats(STUDYID,"-",SUBJID);


    LBSEQ + 1;


    LBTEST = "Mutation Count";
    LBORRES = strip(put('Mutation Count'n,best.));
    LBCAT = "GENOMIC ANALYSIS";
    output;


    LBTEST = "Tumor Purity";
    LBORRES = strip(put('Tumor Purity'n,best.));
    LBCAT = "TUMOR ANALYSIS";
    output;


    LBTEST = "Tumor Mutation Burden";
    LBORRES = strip(put('TMB (Nonsynonymous)'n,best.));
    LBCAT = "GENOMIC ANALYSIS";
    output;


    LBTEST = "Cytolytic Activity Score";
```

```
    LBORRES = strip(put('Cytolytic Activity Score'n,best.));
    LBCAT = "IMMUNE ANALYSIS";
    output;

    keep STUDYID DOMAIN USUBJID SUBJID
         LBSEQ LBTEST LBORRES LBCAT;

    title "LABORATORY DOMAIN";

run;
```

```
proc print data=LB;
run;


/*==========================================================
                LB QC CHECK
==========================================================*/
```

```
proc contents data=lb varnum;
title "LB OVERVIEW";
run;
```

```
proc freq data= LB ;
tables     LBTEST LBORRES LBCAT;
      title"LB DOMAIN FREQUENCY CHECK";
      run;


/*==========================================================
                TUMOR DOMAIN
                        TU
==========================================================*/
```

```
data TU;

    set work.BCMA_RAW_1;

    attrib
        STUDYID length=$20   label="Study Identifier"
        DOMAIN  length=$2    label="Domain Abbreviation"
        USUBJID length=$40   label="Unique Subject Identifier"
        SUBJID  length=$30   label="Subject Identifier"
        TUSEQ   length=8     label="Tumor Identification Sequence Number"
        TUTEST  length=$60   label="Tumor Identification Test Name"
        TUORRES length=$100  label="Tumor Identification Result"
        TUCAT   length=$40   label="Tumor Category";


    STUDYID = "brca_smc_2018";

    DOMAIN = "TU";
```

```
    SUBJID = 'PATIENT ID'n;

    USUBJID = cats(STUDYID,"-",SUBJID);


    TUSEQ + 1;


    TUTEST = "Cancer Type";
    TUORRES = 'CANCER TYPE'n;
    TUCAT = "ONCOLOGY";
    output;


    TUTEST = "Cancer Type Detailed";
    TUORRES = 'CANCER TYPE DETAILED'n;
    TUCAT = "ONCOLOGY";
    output;


    TUTEST = "Oncotree Code";
    TUORRES = 'ONCOTREE CODE'n;
    TUCAT = "TUMOR CLASSIFICATION";
    output;


    TUTEST = "Sample Type";
    TUORRES = 'SAMPLE TYPE'n;
    TUCAT = "TUMOR SAMPLE";
    output;


    TUTEST = "Sample Class";
    TUORRES = 'SAMPLE CLASS'n;
    TUCAT = "TUMOR SAMPLE";
    output;


    TUTEST = "TNM Stage";
    TUORRES = 'TNM Stage'n;
    TUCAT = "CANCER STAGE";
    output;


    keep STUDYID DOMAIN USUBJID SUBJID
         TUSEQ TUTEST TUORRES TUCAT;

    title "TUMOR IDENTIFICATION DOMAIN";

run;

proc print data=TU;
run;

/*=========================================================
```

```
            TU QC CHECK
==========================================================*/


proc contents data= TU varnum;
   title "TU OVERVIEW";
   run;

   proc freq data= TU ;
tables
      TUTEST TUORRES TUCAT;
     title"TU DOMAIN FREQUENCY CHECK";
run;


/*=========================================================
                SUBJECT CHARACTERISTICS
                        SC
==========================================================*/
   data SC;

    set work.BCMA_RAW_1;

    attrib
        STUDYID length=$20   label="Study Identifier"
        DOMAIN  length=$2    label="Domain Abbreviation"
        USUBJID length=$40   label="Unique Subject Identifier"
        SUBJID  length=$30   label="Subject Identifier"
        SCSEQ   length=8     label="Subject Characteristic Sequence Number"
        SCTEST  length=$60   label="Subject Characteristic Test Name"
        SCORRES length=$100  label="Result or Finding in Original Units"
        SCCAT   length=$40   label="Subject Characteristic Category";


    STUDYID = "brca_smc_2018";

    DOMAIN = "SC";

    SUBJID = 'PATIENT ID'n;

    USUBJID = cats(STUDYID,"-",SUBJID);


    SCSEQ + 1;


    SCTEST = "Menopausal Status";

if missing ('MENOPAUSAL STATUS AT DIAGNOSIS'n) then
    SCORRES= "N";
    else
    SCORRES = 'MENOPAUSAL STATUS AT DIAGNOSIS'n;

    SCCAT = "PATIENT CHARACTERISTICS";
    output;
```

```
    SCTEST = "PAM50 Subtype";

if missing('PAM50 subtype'n) then
    SCORRES = "N";
else
    SCORRES = 'PAM50 subtype'n;

SCCAT = "MOLECULAR SUBTYPE";

output;

    SCTEST = "Subtype Consensus";

if missing ('Subtype Consensus'n) then
  SCORRES = "NULL";
  else
    SCORRES = 'Subtype Consensus'n;

    SCCAT = "MOLECULAR CLASSIFICATION";
    output;


    SCTEST = "Sample Class";
if missing ('SAMPLE CLASS'n) then
    SCORRES ="NULL";
    else
    SCORRES = 'SAMPLE CLASS'n;

    SCCAT = "SAMPLE CHARACTERISTICS";
    output;


     keep STUDYID DOMAIN USUBJID SUBJID
         SCSEQ SCTEST SCORRES SCCAT;

    title "SUBJECT CHARACTERISTICS DOMAIN";

run;
```

```
proc print data=SC;
run;

/*=========================================================
             SC QC CHECK
=======================================================*/
```

```
proc contents data= SC varnum;
  title "SC OVERVIEW";
run;
```

```
proc freq data= SC ;
```

```
    tables
        SCTEST SCORRES SCCAT;
    title"TU DOMAIN FREQUENCY CHECK";
run;


/*==========================================================
                SORT DATASETS
==========================================================*/


proc sort data=DM;
    by USUBJID;
run;

proc sort data=LB;
    by USUBJID;
run;

proc sort data=SC;
    by USUBJID;
run;

proc sort data=TU;
    by USUBJID;
run;


/*==========================================================
                CREATE ADAM DATASET
==========================================================*/


data ADAM_MUTATION;

    merge
        DM(in=a)
        LB(in=b)
        SC(in=c)
        TU(in=d);

    by USUBJID;

   if a;



    attrib
        PARAM length=$60    label= "Analysis parameter"
        AVAL  length=$100   label= "Analysis value";


    if LBTEST = "Mutation Count" then do;
        PARAM = "Mutation Count";
        AVAL  = LBORRES;
```

```
        output;
    end;


    else if LBTEST = "TMB" then do;
        PARAM = "Tumor Mutation Burden";
        AVAL  = LBORRES;
        output;
    end;


    else if SCTEST = "PAM50 Subtype" then do;
        PARAM = "PAM50 Subtype";
        AVAL  = SCORRES;
        output;
    end;


    else if TUTEST = "TNM Stage" then do;
        PARAM = "TNM Stage";
        AVAL  = TUORRES;
        output;
    end;


    keep
        STUDYID
        USUBJID
        SUBJID
        SEX
        AGE
        PARAM
        AVAL;

run;
```

```
proc print data=ADAM_MUTATION(obs=50);
run;


/*==========================================================
                ADAM QC CHECK
==========================================================*/
```

```
proc contents data = ADAM_MUTATION;
title "ADAM OVERVIEW";
run;
```

```
proc freq data = adam_mutation;
Tables
       USUBJID
       PARAM
```

```
        AVAL;
title "ADAM FREQUENCY CHECK";
run;
```

```
   proc sql;
select count(*) as total,
count(distinct USUBJID) as duplicate
from ADAM_MUTATION;
quit;


/*==========================================================
        STATISTICAL MUTATION ANALYSIS + VISUALIZATION
==========================================================*/


/*==========================================================
       CREATE NUMERIC ANALYSIS DATASET
==========================================================*/
```

```
data ADAM_NUMERIC;

    set ADAM_MUTATION;


    if PARAM in
    ("Mutation Count",
     "Tumor Mutation Burden");


    AVALN = input(AVAL,best.);

run;


/*==========================================================
     CREATE CHARACTER ANALYSIS DATASET
==========================================================*/
```

```
data ADAM_CHARACTER;

    set ADAM_MUTATION;


    if PARAM in
         ("PAM50 Subtype",
             "TNM Stage");

run;
```

```
proc contents data=ADAM_CHARACTER varnum;
title "ADAM CHARACTER OVERVIEW";
run;
```

```
/*==========================================================
        NUMERIC SUMMARY ANALYSIS
==========================================================*/
```

```
proc means data=ADAM_NUMERIC
mean
median
std
min
max
n
nmiss;

class PARAM;

var AVALN;

title "NUMERIC MUTATION SUMMARY ANALYSIS";

run;
```

```
/*==========================================================
          CATEGORICAL ANALYSIS
==========================================================*/
```

```
proc freq data=ADAM_CHARACTER;

tables
        PARAM
        AVAL
        PARAM*AVAL;

title "CATEGORICAL MUTATION ANALYSIS";

run;
```

```
/*==========================================================
          MUTATION COUNT ANALYSIS
==========================================================*/
```

```
proc means data=BCMA_RAW_1
mean
median
std
min
max;

var 'MUTATION COUNT'n;

title "MUTATION COUNT ANALYSIS";
```

```
run;


/*==========================================================
     TMB ANALYSIS
==========================================================*/
proc means data=BCMA_RAW_1
mean
median
std
min
max;

var 'TMB (NONSYNONYMOUS)'n;

title "TUMOR MUTATION BURDEN ANALYSIS";

run;


/*==========================================================
         PAM50 SUBTYPE ANALYSIS
==========================================================*/
data PAM50_ANALYSIS;

    set BCMA_RAW_1;


    length PAM50_CLEAN $50;


    if missing('PAM50 subtype'n) then
        PAM50_CLEAN = "NULL";

    else
        PAM50_CLEAN = 'PAM50 subtype'n;

run;

/* QC CHECK */
proc freq data=PAM50_ANALYSIS;

tables PAM50_CLEAN;

title "PAM50 SUBTYPE ANALYSIS WITH NULL VALUES";

run;


/*==========================================================
         TNM STAGE ANALYSIS
==========================================================*/
```

```
proc freq data=BCMA_RAW_1;

tables 'TNM Stage'n;

title "TNM STAGE ANALYSIS";

run;


/*===========================================================
       CHEMOTHERAPY ANALYSIS
===========================================================*/

proc means data=BCMA_RAW_1 mean median;

class CHEMOTHERAPY;

var 'MUTATION COUNT'n;

title "CHEMOTHERAPY VS MUTATION COUNT";

run;



/*===========================================================
                TLF CREATION
     TABLES, LISTINGS, FIGURES
===========================================================*/


/*===========================================================
                    TABLE 1
        PAM50 SUBTYPE SUMMARY TABLE
===========================================================*/

proc freq data=PAM50_ANALYSIS noprint;

tables PAM50_CLEAN / out=PAM50_TABLE;

run;


proc report data=PAM50_TABLE nowd;

columns
        PAM50_CLEAN
        COUNT
        PERCENT;

define PAM50_CLEAN / display "PAM50 Subtype";
define COUNT / display "Frequency";
define PERCENT / display "Percentage";
```

```
title "TABLE 1: PAM50 SUBTYPE SUMMARY";

run;


/*==========================================================
                    TABLE 2
          MUTATION COUNT SUMMARY TABLE
==========================================================*/

proc means data=BCMA_RAW_1
n
mean
median
std
min
max
maxdec=2;

var 'MUTATION COUNT'n;

title "TABLE 2: MUTATION COUNT SUMMARY";

run;


/*==========================================================
                    TABLE 3
               TMB SUMMARY TABLE
==========================================================*/

proc means data=BCMA_RAW_1
n
mean
median
std
min
max
maxdec=2;

var 'TMB (NONSYNONYMOUS)'n;

title "TABLE 3: TUMOR MUTATION BURDEN SUMMARY";

run;


/*==========================================================
                    TABLE 4
       CHEMOTHERAPY VS MUTATION COUNT
==========================================================*/

proc means data=BCMA_RAW_1
mean
median
```

```
std
maxdec=2;

class CHEMOTHERAPY;

var 'MUTATION COUNT'n;

title "TABLE 4: CHEMOTHERAPY VS MUTATION COUNT";

run;


/*==========================================================
                    LISTING 1
           SUBJECT LEVEL LISTING
==========================================================*/

proc print data=ADAM_MUTATION(obs=50);

var
USUBJID
SUBJID
SEX
AGE
PARAM
AVAL;

title "LISTING 1: SUBJECT LEVEL MUTATION ANALYSIS";

run;


/*==========================================================
                    LISTING 2
             PAM50 SUBJECT LISTING
==========================================================*/

proc print data=PAM50_ANALYSIS(obs=50);

var
'Patient ID'n
PAM50_CLEAN
'TNM Stage'n
'MUTATION COUNT'n;

title "LISTING 2: PAM50 SUBTYPE SUBJECT LISTING";

run;


/*==========================================================
                    FIGURE 1
             PAM50 BAR CHART
==========================================================*/
```

```
proc sgplot data=PAM50_ANALYSIS;

vbar PAM50_CLEAN;

title "FIGURE 1: PAM50 SUBTYPE DISTRIBUTION";

run;


/*==========================================================
                    FIGURE 2
          MUTATION COUNT HISTOGRAM
==========================================================*/
```

```
proc sgplot data=BCMA_RAW_1;

histogram 'MUTATION COUNT'n;

density 'MUTATION COUNT'n;

title "FIGURE 2: MUTATION COUNT DISTRIBUTION";

run;


/*==========================================================
                    FIGURE 3
             PAM50 BOXPLOT
==========================================================*/
```

```
proc sgplot data=BCMA_RAW_1;

vbox 'MUTATION COUNT'n /
category='PAM50 subtype'n;

title "FIGURE 3: MUTATION COUNT BY PAM50";

run;


/*==========================================================
                    FIGURE 4
           TMB VS MUTATION SCATTER
==========================================================*/
```

```
proc sgplot data=BCMA_RAW_1;

scatter
x='TMB (NONSYNONYMOUS)'n
y='MUTATION COUNT'n;

title "FIGURE 4: TMB VS MUTATION COUNT";

run;
```

```
/*==========================================================
                   QC CHECK
==========================================================*/

proc contents data=ADAM_MUTATION varnum;

title "TLF QC CHECK";

run;


/*=========================================================
               EXPORT PDF REPORT
=========================================================*/

ods pdf file="/home/u64383806/TLF_REPORT.pdf";



/*=========================================================
               REPORT TITLE
=========================================================*/

title "Clinical Mutation Analysis Report";


/*=========================================================
               TABLE 1
=========================================================*/

proc freq data=SC;

tables SCORRES;

title "TABLE 1 - PAM50 SUBTYPE SUMMARY";

run;


/*=========================================================
               TABLE 2
=========================================================*/

proc means
data=BCMA_RAW_1
n
mean
median
std
min
max
maxdec=2;

var 'Mutation Count'n;
```

```
title "TABLE 2 - MUTATION COUNT SUMMARY";

run;


/*==========================================================
                TABLE 3
==========================================================*/

proc means
data=BCMA_RAW_1
n
mean
median
std
min
max
maxdec=2;

var 'TMB (Nonsynonymous)'n;

title "TABLE 3 - TMB SUMMARY";

run;


/*==========================================================
                LISTING 1
==========================================================*/

proc print data=ADAM_MUTATION(obs=30);

var
    USUBJID
    SEX
    AGE
    PARAM
    AVAL;

title "LISTING 1 - SUBJECT LEVEL LISTING";

run;


/*==========================================================
                FIGURE 1
==========================================================*/

proc sgplot data=SC;

vbar SCORRES;

title "FIGURE 1 - PAM50 BAR CHART";
```

```
run;
```